

SKILL WEEK 2

1. To Analyze process abstraction, execute `fork()`, understand `exec()` family, analyze parent-child relationships, inspect process tree, practice system call tracing.

Task-1: Using a manual command understand the fork() and exec() system calls

1. Inspecting the Process Tree (pstree)

```
imvarls@LAPTOP-VQ4D70L7:~$ pstree -p
systemd(1)─agetty(213)
          └─chronyd-starter(108)─chronyd(209)─chronyd(226)
          └─cron(113)
          └─dbus-daemon(117)
          └─init-systemd(Ub(2))─SessionLeader(1157)─Relay(1162)(1159)─bash(1162)─pstree(1216)
                                     └─init(7)─{init}(8)
                                     └─login(322)─bash(395)
                                     └─{init-systemd(Ub)}(9)
          └─networkd-dispat(154)
          └─rsyslogd(185)─{rsyslogd}(234)
                          └─{rsyslogd}(235)
                          └─{rsyslogd}(236)
          └─systemd(364)─(sd-pam)(368)
          └─systemd-journal(47)
          └─systemd-logind(158)
          └─systemd-resolve(80)
          └─systemd-udevd(90)─(udev-worker)(1209)
                              └─(udev-worker)(1210)
          └─unattended-upgr(270)─{unattended-upgr}(284)
          └─wsl-pro-service(1183)─{wsl-pro-service}(1184)
                                   └─{wsl-pro-service}(1185)
                                   └─{wsl-pro-service}(1186)
                                   └─{wsl-pro-service}(1187)
                                   └─{wsl-pro-service}(1188)
                                   └─{wsl-pro-service}(1189)
                                   └─{wsl-pro-service}(1190)

imvarls@LAPTOP-VQ4D70L7:~$
```

2. System Call Tracing (strace)

[illegible]

```

jetrandon("\x40\x80\x33\xc4\x8d\xb3\x2f\x79", 8, GND_NONBLOCK) = 0
munmap(0x785918299800, 20887) = 0
statfs(</sys/fs/selinux>, {f_type=SYSFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0, f_bavail=0, f_files=0, f_ffree=0, f_fsid={val={0xd2891b6d, 0x1950fa38}}, f_namelen=255, f_frsize=4096, f_flags=ST_VALID|ST_NOSUID|ST_NOEXEC|ST_NO
statfs(</selinux>, 0x7ff6b9a36a00) = -1 ENOENT (No such file or directory)
rsh(NULL) = 0x5c24972c4000
rsh(0x5c24972c4000) = 0x5c24972c4000
openat(AT_FDCWD, "/proc/filesystems", 0_RDONLY|O_CLOEXEC) = 3
stat(<, {st_mode=1_FREG(0444, st_size=0, ...)} = 0
read(<, "nodev(</sys/fs/inodev/tmpfs/inodev/libd..., 1024) = 442
ioctl(<, 0, SEEK_CUR) = 442
ioctl(<, 0, SEEK_SET) = 416
close(<) = 0
openat(AT_FDCWD, "/proc/mounts", 0_RDONLY|O_CLOEXEC) = 3
stat(<, {st_mode=1_FREG(0444, st_size=0, ...)} = 0
read(<, "none /usr/lib/modules/6.10.11.2-..., 1024) = 1824
read(<, "0 0 0/devpts /dev/pts devpts rw ..., 1024) = 1824
read(<, "access=client,msize=65536,trans..., 1024) = 1824
read(<, "780.noswap 0 0/ntmpfs /run/user/1-..., 1024) = 250
read(<, "", 1024) = 0
close(<) = 0
access(</etc/selinux/config>, F_OK) = -1 ENOENT (No such file or directory)
fcntl(<, F_GETFD) = 0
fcntl(<, F_GETFD) = 0
fcntl(<, F_GETFD) = 0
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
poll([fd=0, events=0], [fd=0, events=0], [fd=0, events=0], 3, 0) = 0 (Timeout)
rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x78590fe45cb0}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
openat(AT_FDCWD, "/proc/self/maps", 0_RDONLY|O_CLOEXEC) = 3
rlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INF|RTITVT}) = 0
stat(<, {st_mode=1_FREG(0444, st_size=0, ...)} = 0
read(<, "Sc2edd762800-Sc2edd854800 r--p 0-..., 1024) = 1824
read(<, "30 13647 /u-..., 1024) = 1824
read(<, "30 13650 /-..., 1024) = 1824
read(<, " /usr/lib/x86_64-linux-gn-..., 1024) = 1824
read(<, "uu /usr/lib-..., 1024) = 661
close(<) = 0
sched_getaffinity(1238, 32, [0 1 2 3 4 5 6 7 8 9 10 11]) = 0
rt_sigaction(SIGSEGV, {sa_handler=0x5c2adda878c0, sa_mask=[], sa_flags=0}, 0) = 0
sigaltstack(NULL, {ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=0}) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x785918299800
protect(0x785918299800, 4096, PROT_NONE) = 0
sigaltstack({ss_sp=0x785918299800, ss_flags=0, ss_size=8192}, NULL) = 0
jettid() = 1238
rt_sigaction(SIGSEGV, {sa_handler=0x5c2adda878c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x78590fe45cb0}, NULL, 0) = 0
rt_sigaction(SIGBUS, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
rt_sigaction(SIGBUS, {sa_handler=0x5c2adda878c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x78590fe45cb0}, NULL, 0) = 0
prctl(PR_GET_AUXV, 0x7ff6b9a36a00, 0x200, 0, 0) = 440
statx(AT_FDCWD, "/usr/share/coreutils/locales/ls", AT_STATX_SYNC_AS_STAT, STATX_ALL, [str_mask=STATX_ALL|STATX_MNT_ID, str_attributes=0, stx_mode=0_IFDIR(0755, stx_size=4096, ...)] = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/umcore", AT_STATX_SYNC_AS_STAT, STATX_ALL, [str_mask=STATX_ALL|STATX_MNT_ID, str_attributes=0, stx_mode=0_IFDIR(0755, stx_size=4096, ...)] = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/umcore/en-US.fel", 0_RDONLY|O_CLOEXEC) = 3
statx(<, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, [str_mask=STATX_ALL|STATX_MNT_ID, str_attributes=0, stx_mode=0_IFREG(0444, stx_size=4080, ...)] = 0
read(<, "B Common strings shared across a-..., 4080) = 4080
read(<, "", 32) = 0
close(<) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/ls", AT_STATX_SYNC_AS_STAT, STATX_ALL, [str_mask=STATX_ALL|STATX_MNT_ID, str_attributes=0, stx_mode=0_IFDIR(0755, stx_size=4096, ...)] = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/ls/en-US.fel", 0_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rsh(0x5c2497386000) = 0x5c2497386000
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x78590fe45cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x78590fe45cb0}, 0) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x78590fe45cb0}, {sa_handler=0x5c2adda878c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x78590fe45cb0}, 0) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x78590fe45cb0}, {sa_handler=0x5c2adda878c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x78590fe45cb0}, 0) = 0
ioctl(<, TCGETS2, {c_iflag=ICNL|IXON, c_oflag=NL0|CR|TAB|BS|VT0|FF|DPOST|ONLCR, c_cflag=018400|CS|CREAD, c_lflag=ISIC|CANON|ECHN|ECHN|ECHO|ECHO|IEXTEN|ECHOCTL|ECHONK, ...}) = 0
rsh(0x5c2497327000) = 0x5c2497327000
ioctl(<, TCGETS2, {c_iflag=ICNL|IXON, c_oflag=NL0|CR|TAB|BS|VT0|FF|DPOST|ONLCR, c_cflag=018400|CS|CREAD, c_lflag=ISIC|CANON|ECHN|ECHN|ECHO|ECHO|IEXTEN|ECHOCTL|ECHONK, ...}) = 0
ioctl(<, TIOCGWINSZ, {ws_row=10, ws_col=120, ws_xpixel=0, ws_ypixel=0}) = 0
ioctl(<, TIOCGWINSZ, {ws_row=10, ws_col=120, ws_xpixel=0, ws_ypixel=0}) = 0
ioctl(<, TCGETS2, {c_iflag=ICNL|IXON, c_oflag=NL0|CR|TAB|BS|VT0|FF|DPOST|ONLCR, c_cflag=018400|CS|CREAD, c_lflag=ISIC|CANON|ECHN|ECHN|ECHO|ECHO|IEXTEN|ECHOCTL|ECHONK, ...}) = 0
ioctl(<, TCGETS2, {c_iflag=ICNL|IXON, c_oflag=NL0|CR|TAB|BS|VT0|FF|DPOST|ONLCR, c_cflag=018400|CS|CREAD, c_lflag=ISIC|CANON|ECHN|ECHN|ECHO|ECHO|IEXTEN|ECHOCTL|ECHONK, ...}) = 0
statx(AT_FDCWD, "", AT_STATX_SYNC_AS_STAT, STATX_ALL, [str_mask=STATX_ALL|STATX_MNT_ID, str_attributes=0, stx_mode=0_IFDIR(0755, stx_size=4096, ...)] = 0
statx(AT_FDCWD, "", AT_STATX_SYNC_AS_STAT, STATX_ALL, [str_mask=STATX_ALL|STATX_MNT_ID, str_attributes=0, stx_mode=0_IFDIR(0755, stx_size=4096, ...)] = 0
openat(AT_FDCWD, "", 0_RDONLY|O_NONBLOCK|O_CLOEXEC|O_DIRECTORY) = 3
stat(<, {st_mode=0_IFDIR(0755, stx_size=4096, ...)} = 0
newfstatat(AT_FDCWD, "", {st_mode=0_IFDIR(0755, stx_size=4096, ...)}, 0) = 0
jetdents64(<, 0x5c249724f8150 /* 30 entries */, 32768) = 944
jetdents64(<, 0x5c249724f8150 /* 0 entries */, 32768) = 0
close(<) = 0
write(<, "File.txt File.txt O SSP copy-..., 209File.txt File.txt O SSP copy.c file filename filename.c.save first.c.save.1 process.c task2.c
File.tx File.txt.c copy copy.file.c file.c filename.c first.c.save process task2
) = 209
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x785918299800, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++
tavaris@LAPTOP-vq4d7m1y:~$

```

Task-2: Write a C program to create a new process and accepting the user commands using fork() and exec() system calls.

```
imvarls@LAPTOP-VQ4D70L7:~$ nano process.c
```

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    // Create a new process
    pid = fork();

    if (pid < 0) {
        // Error handling if fork fails
        perror("Fork failed");
        exit(1);
    }
    else if (pid == 0) {
        // --- Child Process ---
        printf("--> [Child Process]\n");
        printf("    Child PID: %d\n", getpid());
        printf("    Parent PID: %d\n", getppid());

        // Define an argument list for the command you want to execute
        // Example: executing "ls -l"
        char *args[] = {"ls", "-l", NULL};

        printf("    Executing 'ls -l' using execvp()...\n\n");

        // execvp replaces the current child process image with a new process image
        execvp(args[0], args);

        // If execvp returns, an error occurred
        perror("execvp failed");
        exit(1);
    }
}
```

```

    }
    else {
        // --- Parent Process ---
        printf("--> [Parent Process]\n");
        printf("    Parent PID: %d\n", getpid());
        printf("    Child PID created: %d\n", pid);

        // Parent waits for the child process to complete execution
        int status;
        wait(&status);

        printf("\n--> [Parent Process] Child process has finished. Parent exiting.\n");
    }

    return 0;
}

```

```

imvarls@LAPTOP-VQ4D70L7:~$ nano process.c
imvarls@LAPTOP-VQ4D70L7:~$ gcc process.c -o process
imvarls@LAPTOP-VQ4D70L7:~$ ./process|

```

```

imvarls@LAPTOP-VQ4D70L7:~$ nano process.c
imvarls@LAPTOP-VQ4D70L7:~$ nano process.c
imvarls@LAPTOP-VQ4D70L7:~$ gcc process.c -o process
imvarls@LAPTOP-VQ4D70L7:~$ ./process
--> [Parent Process]
    Parent PID: 1568
    Child PID created: 1569
--> [Child Process]
    Child PID: 1569
    Parent PID: 1568
    Executing 'ls -l' using execvp(...)

total 152
-rwxr-xr-x  1 imvarls imvarls 16256 Aug  4 07:11 File.txt
-rw-r--r--  1 imvarls imvarls   78 Aug  4 15:19 File1.tx
-rw-r--r--  1 imvarls imvarls   54 Aug  4 15:20 File1.txt
-rw-r--r--  1 imvarls imvarls  1308 Aug  4 07:11 File1.txt.c
drwxr-xr-x 11 imvarls imvarls  4096 Jul 30 09:22 OSSP
-rwxr-xr-x  1 imvarls imvarls 16200 Aug  4 15:20 copy
-rw-r--r--  1 imvarls imvarls   820 Aug  4 15:20 copy.c
-rw-r--r--  1 imvarls imvarls  1529 Aug  4 14:53 copy_file.c
-rwxr-xr-x  1 imvarls imvarls 16512 Aug  4 15:35 file
-rw-r--r--  1 imvarls imvarls  2251 Aug  4 15:35 file.c
-rwxr-xr-x  1 imvarls imvarls 16224 Aug  4 15:01 filename
-rw-r--r--  1 imvarls imvarls   944 Aug  4 15:01 filename.c
-rw-----  1 imvarls imvarls     6 Jul 25 04:19 filename.c.save
-rw-----  1 imvarls imvarls    23 Jul 28 06:54 first.c.save
-rw-----  1 imvarls imvarls   144 Jul 28 07:09 first.c.save.1
-rwxr-xr-x  1 imvarls imvarls 16344 Aug  4 16:16 process
-rw-r--r--  1 imvarls imvarls  1399 Aug  4 16:15 process.c
-rwxr-xr-x  1 imvarls imvarls 16256 Aug  4 07:06 task2
-rw-r--r--  1 imvarls imvarls  1308 Aug  4 07:05 task2.c

--> [Parent Process] Child process has finished. Parent exiting.
imvarls@LAPTOP-VQ4D70L7:~$ |

```